5. Friday, June 28. The box and product topologies (§ 15, 19)

Last Time: • Arbitrary product of sets $\prod_{i\in\Lambda} X_i$

• If each $(X_i, T_i)$ is a Top. space Then $\mathcal{B}' = \left\{ \prod_{i\in\Lambda} U_i \mid U_i \in T_i \right\}$ is a basis for a Topology.

Definition 1: The Topology $T(\mathcal{B}')$ is called The box Topology.

Example: If $\{(X_i, T_i) \mid i\in\Lambda\}$ is a family of discrete spaces, Then The box Topology on $\prod_{i\in\Lambda} X_i$ is The discrete Topology. Just need to prove That singletons are open in $\prod_{i\in\Lambda} X_i$
Given $a \in \prod_{i\in\Lambda} X_i$, we have $\{a\} = \prod_{i\in\Lambda} \{a(i)\}$, and $\{a(i)\} \subseteq X_i$ is open because $X_i$ is discrete
Hence $\{a\}$ is open and $\prod_{i\in\Lambda} X_i$ is discrete.

Open sets in The box Topology can be Too small if There are infinitely many sets!

Example: For each $n \in \mathbb{N}$, Consider $(X_i, T_i) = \mathbb{R}_{std}$. Consider $\prod_{i\in\mathbb{N}} \mathbb{R}_{std}$ with The box Topology.
We have an open set $\left(-\frac{1}{i}, \frac{1}{i}\right) \subseteq X_i$. Hence $U := \prod_{i\in\mathbb{N}} \left(-\frac{1}{i}, \frac{1}{i}\right)$ is open wrt The box Topology
We will see later (or next class) why This open is Too small.

**Definition 2**: Given a collection of sets $\{X_i \mid i \in \Lambda\}$, and $j \in \Lambda$. The **j-Th projection** (or projection to the j-Th coordinate) is the function

$$p_j : \prod_{i \in \Lambda} X_i \longrightarrow X_j, \qquad p_j(a) = a(j)$$

Example: If $\Lambda = \{1, \dots, n\}$ Then $p_j(a_1, \dots, a_n) = a_j$

**Note**: If $U_j \subseteq X_j$, Then $p_j^{-1}(U_j) = \left\{ a \in \prod_{i \in \Lambda} X_i \mid a(j) \in U_j \right\} = U_j \times \prod_{i \in \Lambda - \{j\}} X_i$

Hence, a generic element $\prod_{i \in \Lambda} U_i$ of the box basis can be written as

$\prod_{i \in \Lambda} U_i = \bigcap_{i \in \Lambda} p_i^{-1}(U_i)$. This intersection can be infinite, hence too small.

**Definition/Lemma 3**: Let $\{(X_i, T_i) \mid i \in \Lambda\}$ be a family of topological spaces. Then the collection $S = \left\{ p_j^{-1}(U_j) \subseteq \prod_{i \in \Lambda} X_i \mid j \in \Lambda, U_j \in T_j \right\}$ is a subbasis for a topology on $\prod_{i \in \Lambda} X_i$, called the **product topology**.

**Proof:** To show $\mathcal{S}$ is a subbasis, just need to show $\prod_{i\in\Lambda} X_i = \bigcup_{S\in\mathcal{S}} S$, but that's clear because $\prod_{i\in\Lambda} X_i \in \mathcal{S}$. (Fix your favorite $j\in\Lambda$. Then $X_j \subset T_j$ and $\prod_{i\in\Lambda} X_i = \pi_j^{-1}(X_j)$.) □

**Note:** Recall that $\pi_j^{-1}(U_j) = U_j \times \prod_{i\in\Lambda\setminus\{j\}} X_j$. The basis associated to $\mathcal{S}$ is

$\mathcal{B}_{\mathcal{S}} = \{$ finite intersections of sets in $\mathcal{S}\}$ "The product basis"

$= \left\{ \bigcap_{j\in J} \pi_j^{-1}(U_j) \mid J\subset\Lambda \text{ is finite}, U_j\in T_j \ \forall j\in J \right\}$

$= \left\{ \prod_{j\in J} U_j \times \prod_{i\in\Lambda-J} X_j \mid J\subseteq\Lambda \text{ finite}, U_j\in T_j \ \forall j\in J \right\}$

$= \left\{ \prod_{i\in\Lambda} U_i \in \mathcal{B}' \mid U_i \neq X_i \text{ only for finitely many } i\in\Lambda \right\}$.

**Consequence:** If $\Lambda$ is finite, Then $\mathcal{B}_{\mathcal{S}} = \mathcal{B}'$ (The product and box bases) Hence The product and the box Topology coincide.

Note: If $\Lambda$ is infinite and $\{X_i \mid i \in \Lambda\}$ is a family of discrete spaces, Then The product Topology on $\prod_{i \in \Lambda} X_i$ is not discrete in general. (pset 2).

Fact: If $\mathcal{B}_i$ is a basis for $(X_i, T_i)$ Then

$$\left\{ \prod_{i \in \Lambda} U_i \subseteq \prod_{i \in \Lambda} X_i \;\middle|\; U_i \in \mathcal{B}_i \;\; \forall i \in \Lambda \right\} \text{ is a basis for The box Topology}$$

Example: Consider $\mathbb{R}^n = \prod_{i=1}^{n} \mathbb{R}$. with The product Topology (= box Topology)

Then The "open rectangles" $(a_1, b_1) \times (a_2, b_2) \times \dots \times (a_n, b_n)$ form a basis.

We will come back To Topologies on $\mathbb{R}^n$ when we study metric spaces

# Continuous maps (§ 18)

**Definition 4:** Let $X$ and $Y$ be two topological spaces. A function $f: A \to B$ is **continuous** if for every open set $U \subseteq Y$, the preimage $f^{-1}(U) \subseteq X$ is open.

**Examples:**

1) Let $(X, T)$ be a topological space. Then the identity map $I_X: X \to X$ where $I_X(a) = a \ \forall a$ is continuous. In fact, if $U \subseteq X$ is open, then $I_X^{-1}(U) = U$ is also open.

2) More generally, if $T_1$ and $T_2$ are topologies on $X$, then $I_X: (X, T_1) \to (X, T_2)$ is continuous if and only if $T_1$ is finer than $T_2$.

In fact, that map is continuous iff for all $U \in T_2$, we have $I_X^{-1}(U) \left(= U\right) \in T_1$.

iff $T_2 \subseteq T_1$.